

Pair correlations survive locally in stripe states

Complete retrospective measurements and square A/B results

Ladder MPS+MF project

22 September 2026

Main result. Stripe formation does not eliminate short-distance singlet-pair correlations. In the representative square, cubic and trellis stripe snapshots, these correlations concentrate near hole-rich magnetic domain walls, while propagation over 16–24 rungs is strongly suppressed. Rung–leg correlations retain a predominantly negative relative sign. This separates local pairing from long-distance order, but does not establish fluctuating superconductivity, phase stiffness or a dynamical phase-disordering mechanism.

New square result. All four square two-ladder runs at $t_0 = 1.4$ and $V = -0.4, -0.2$ pass the stored fixed-point gates. Both seeds approach nearly identical paired A/B profiles and pair correlations at each V , despite the displaced stripe component in the initial conditions. This supports survival of the paired solution in the tested square cell. It does not establish stability against arbitrary transverse periods or a global minimum.

1 Coverage and operator convention

The synchronized retrospective campaign now covers all 56 branches and 58 spatial MPSs: 49 receipts/51 MPSs in retry1 and seven receipts/MPSs in retry2. The new square A/B campaign supplies eight further diagnostics from four accepted branches. The isolated reference is the earlier $L = 64$, $t_0 = 1.4$, $V = 0$, $\chi = 1200$ backbone. All coupled measurements use $\chi = 200$, $U = 8$, $n = 15/16$, and $L = 64$ rungs per MPS. Retrospective endpoints remain *unaccepted maximum-iteration snapshots*; completing measurements does not change their scientific convergence status. Trellis A/B labels are spatial ladders, not temporal cycle phases.

Use the stored, unnormalized bond singlet

$$D_{ab} = c_{a\uparrow}c_{b\downarrow} - c_{a\downarrow}c_{b\uparrow}, \quad P_{ab,cd} = \langle D_{ab}^\dagger D_{cd} \rangle, \quad P_{ab,cd}^c = P_{ab,cd} - \langle D_{ab} \rangle^* \langle D_{cd} \rangle. \quad (1)$$

The manuscript’s normalized singlet is $\Delta = D/\sqrt{2}$; its correlations are one half of the values here. Stored onsite operators are instead $c_\uparrow c_\downarrow$. The 318-coordinate basis comprises 64 rung, 126 nearest-neighbor leg and 128 onsite operators. Rung–leg separation is defined by the start rung, so bond midpoints differ by half a rung.

The older isolated-ladder file stores within-class Hermitian-field covariances, with non-onsite operators divided by $\sqrt{2}$. On this real, number-conserving reference and for *disjoint* bonds, that covariance equals $\text{Re}\langle D_i^\dagger D_j \rangle$: $(\langle D_i D_j^\dagger \rangle + \langle D_i^\dagger D_j \rangle)/2$. The anomalous expectation is zero. Contact rung entries and overlapping leg entries are excluded. Rung–leg cross-channel data are unavailable for this older reference and are not reconstructed. Its larger bond dimension precludes a controlled attribution of all differences to transverse coupling.

2 Short-distance survival and long-distance suppression

Distance curves average $|P_{ij}|$ or $|P_{ij}^c|$ over both orientations with rung bonds in rungs 9–56 and leg-bond starts in 9–55, so both endpoints remain in that bulk window. The signed averages and their spatial 10th/90th percentiles are saved separately. Spatial spread is not a statistical confidence interval. Table 1 averages the entries in two separation bands; different separations carry the number of available bulk pairs.

Table 1: Rung-pair magnitudes in the common stored convention. Representative paired-seed endpoints; trellis two-ladder entry is A. All coupled rows in this table are unaccepted snapshots. S and L denote separations 2–4 and 16–24.

State (t_0, V)	$\overline{ P^c }_S$	$\overline{ P^c }_L$	$\overline{ P }_L$
Isolated (1.4, 0), $\chi = 1200$	$2.0345 \cdot 10^{-2}$	$5.2195 \cdot 10^{-3}$	$5.2195 \cdot 10^{-3}$
Square paired (1.4, -0.4)	$1.1816 \cdot 10^{-2}$	$3.0361 \cdot 10^{-4}$	$1.1661 \cdot 10^{-2}$
Square stripe (1.4, 0)	$1.5090 \cdot 10^{-2}$	$9.8143 \cdot 10^{-5}$	$9.8143 \cdot 10^{-5}$
Cubic stripe (1.4, -0.4)	$1.0064 \cdot 10^{-2}$	$1.2935 \cdot 10^{-7}$	$1.2935 \cdot 10^{-7}$
Trellis one-ladder paired (1, 0)	$5.9777 \cdot 10^{-3}$	$4.6411 \cdot 10^{-5}$	$4.3956 \cdot 10^{-3}$
Trellis two-ladder stripe (1, 0)	$5.8648 \cdot 10^{-3}$	$6.6518 \cdot 10^{-7}$	$6.6518 \cdot 10^{-7}$

The square $V = 0$ stripe retains roughly 74% of the isolated reference’s short-distance magnitude but only 1.9% of its long-distance magnitude. The latter is a descriptive comparison across $\chi = 200$ and 1200, not a bond-dimension-controlled suppression factor. Both square stripe seeds give the same qualitative result ($8.19\text{--}9.81 \cdot 10^{-5}$ at long distance).

Comparisons at matched intraladder parameters are sharper. Square paired and cubic striped snapshots at (1.4, -0.4) have comparable short-distance connected magnitudes, while the cubic long-distance value is about $2.2\text{--}2.3 \cdot 10^3$ smaller. For trellis at (1, 0), the one-/two-ladder short-distance values differ by only a few percent, whereas the striped two-ladder long-distance connected value is about 70 times smaller. These are differences between saved endpoints and spatial embeddings, not a controlled causal intervention or a stationary phase ranking.

In the square paired state, approximately 97.4% of the long-distance *signed* full rung correlation is the disconnected product of anomalous expectations. The corresponding raw/connected curves are essentially identical in the representative stripes. Thus an ordered-state plateau and the decay of residual pair fluctuations answer different questions.

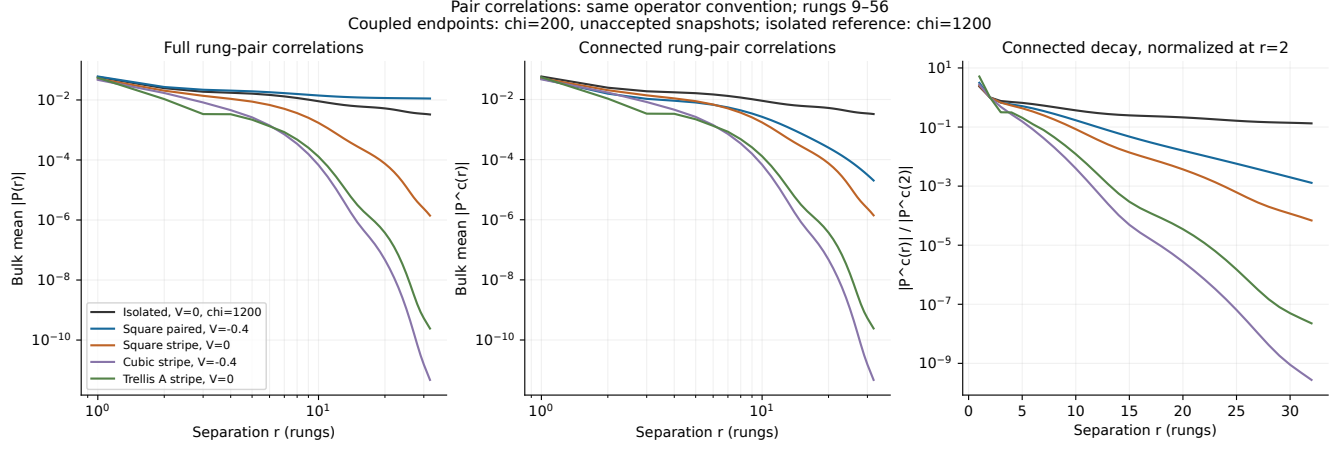


Figure 1: Full, connected and normalized connected rung-pair correlations. The trellis curve has $t_0 = 1$, $V = 0$; the other four have $t_0 = 1.4$. Magnitudes retain oscillatory contributions rather than cancelling signs. The farthest, very small tails have no independent truncation-error estimate.

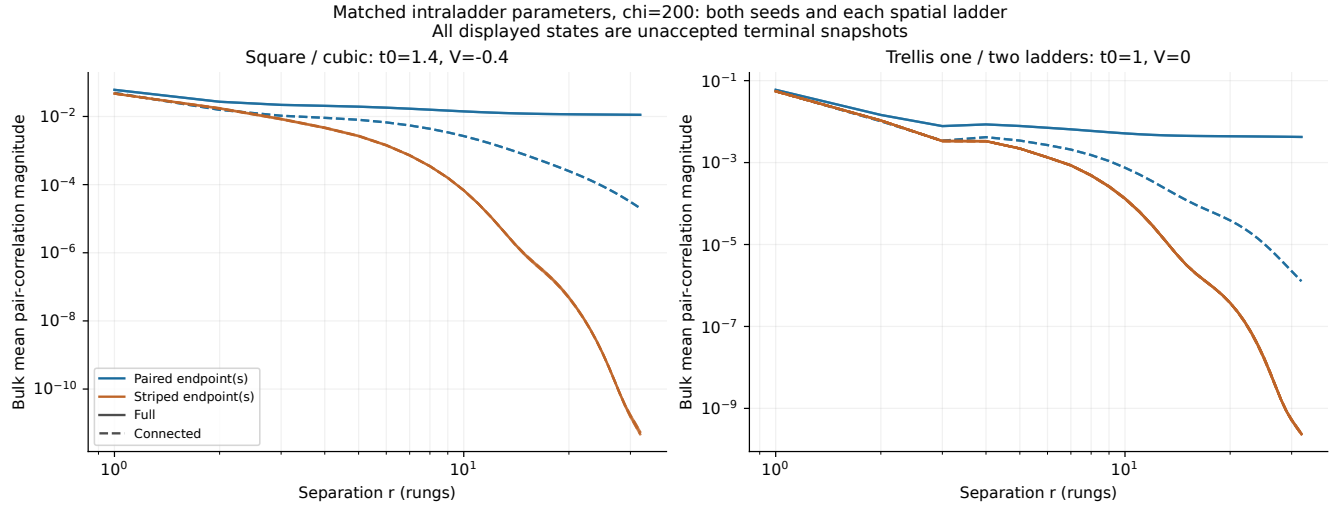


Figure 2: Matched intraladder-parameter comparisons including both initial families and, where present, both spatial ladders. Solid/dashed lines separate full and connected values. Similar short-distance amplitudes coexist with strongly different long-distance behavior.

3 Bond signs and location within stripes

For the representative square paired, square striped and cubic striped states, rung-rung and leg-leg averages are positive while rung-leg averages are negative. Over separations 2–8 in the bulk, the selected square stripe gives leg-leg $4.31 \cdot 10^{-3}$ and rung-leg $-7.07 \cdot 10^{-3}$; the cubic stripe gives $1.90 \cdot 10^{-3}$ and $-3.66 \cdot 10^{-3}$, respectively. These are connected values. Across all 58 retrospective MPSs, the negative fraction of connected rung-leg0 entries in this band is 96.3–100% on square, 87.2–100% on cubic and 100% on trellis. Leg1 is retained separately in the data table. This is compatible with a local rung/leg sign structure commonly called *d*-wave-like, not proof of a condensate or pair-density wave.

The trellis case also warns against an overly uniform form-factor claim: its leg-leg correlations change sign with separation. In the two-ladder striped example, the signed leg0 mean over 2–8 is about $-4.1 \cdot 10^{-5}$ even though the nearest values are positive and rung-leg entries remain negative. The signed channel plots and tables should accompany magnitude plots whenever internal pair structure is discussed.

Define a local diagnostic

$$A(i) = \frac{1}{6} \sum_{r=2}^4 (|P_{\text{rung}}^c(i, i+r)| + |P_{\text{rung}}^c(i, i-r)|). \quad (2)$$

Use centers 13–52, so all six neighbors lie in the common bulk window. Compare the ten centers in the highest hole-density quartile with the ten in the lowest; no fitting or peak tuning is needed for this amplitude ratio. The ratio is 5.85–6.68 for the two square $V = 0$ seeds, 19.39–19.59 for the two cubic $(1.4, -0.4)$ seeds, and 5.29–5.36 over both seeds/ladders of striped trellis. Interior hole-density maxima lie within half a rung of the discrete staggered-spin sign crossings. The strongest short-distance pair correlations therefore sit near hole-rich antiphase walls. Spatial observations in a single texture are correlated; these ratios are descriptive and do not establish a mechanism of stripe-induced attraction.

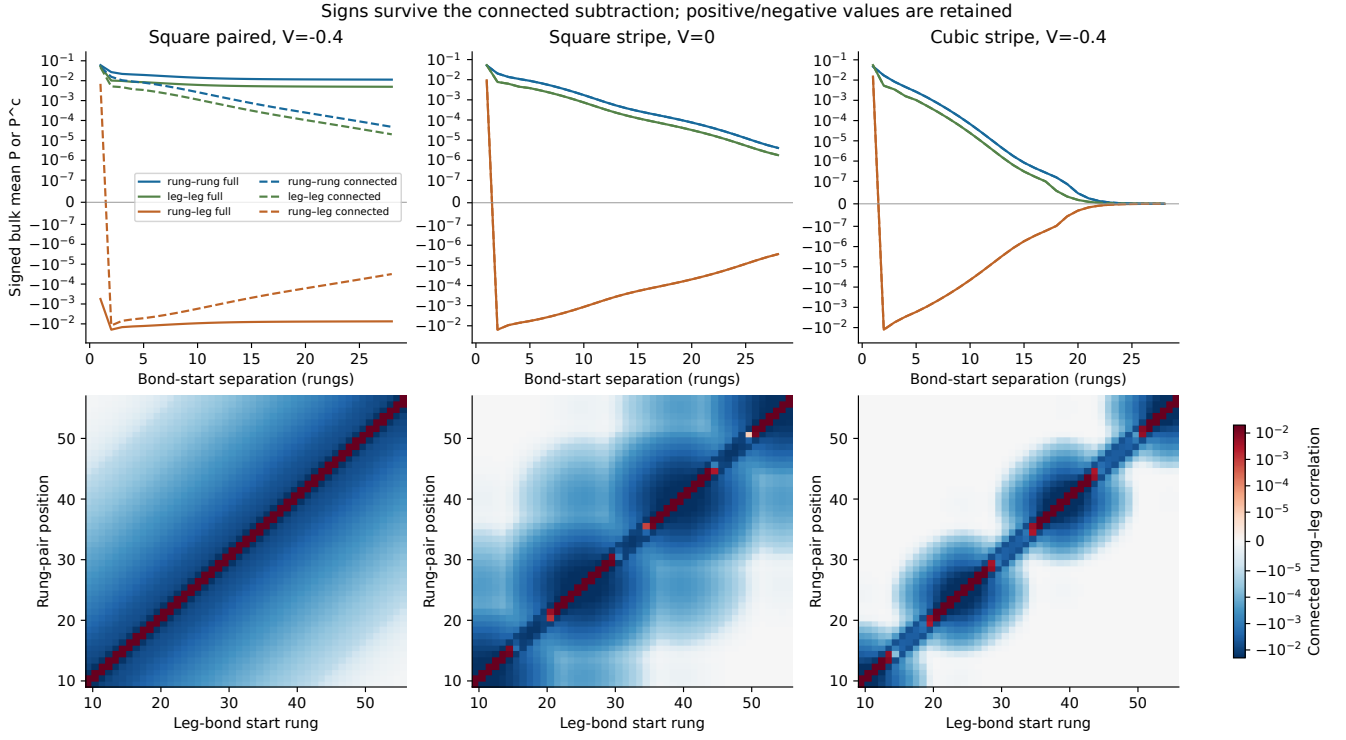


Figure 3: Signed full/connected rung-rung, leg0-leg0 and rung-leg0 curves, with connected rung-leg maps below. Symmetric-log scales preserve sign and small values; matrix contact regions should not be interpreted as propagation. The isolated reference lacks cross-channel measurements.

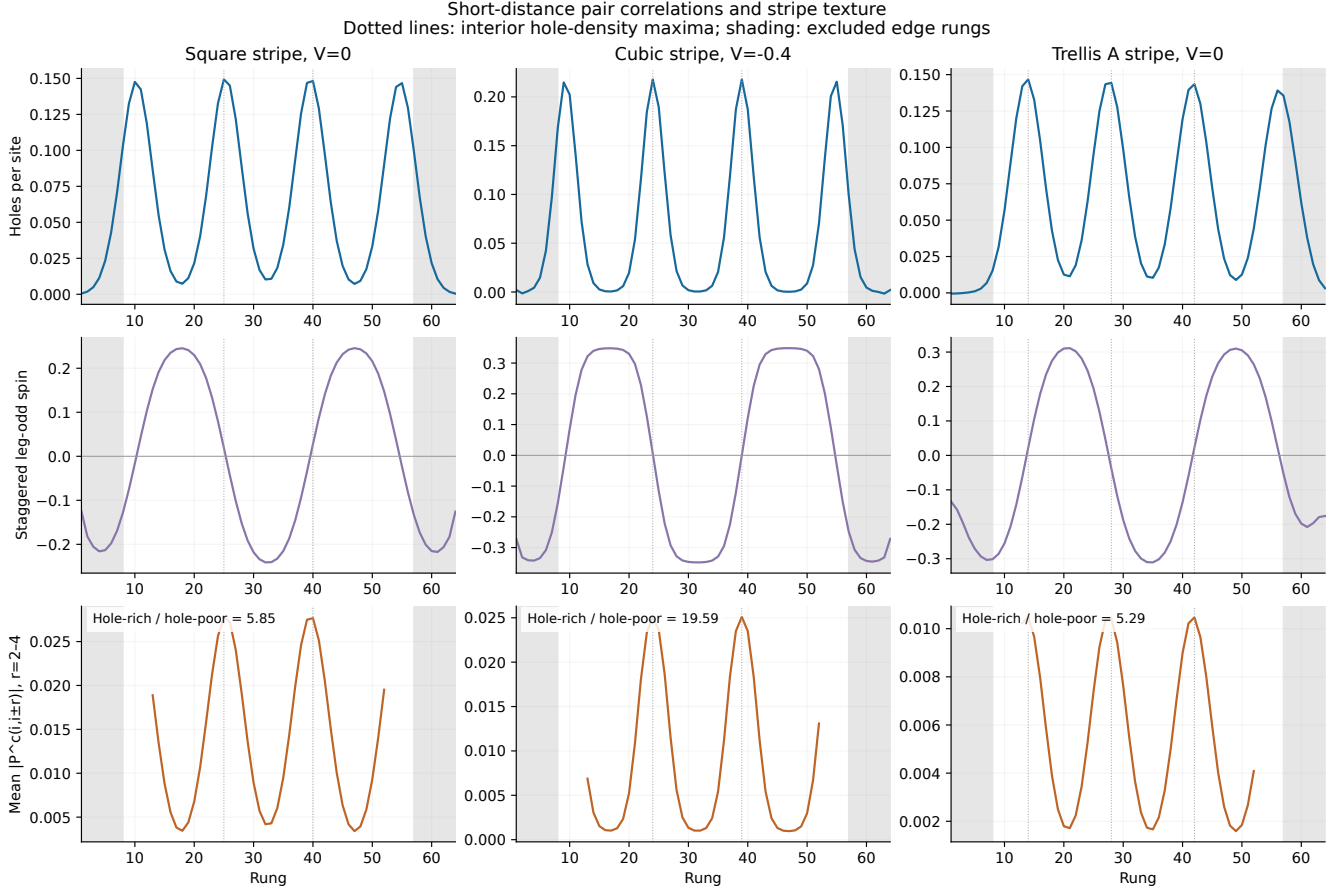


Figure 4: Hole density, staggered leg-odd spin and local pair diagnostic $A(i)$. Shading excludes the outer eight rungs; $A(i)$ uses the stricter center window 13–52. Dotted lines identify interior hole maxima. The three columns are representative snapshots, not three independent statistical ensembles.

4 Reference, window and boundary sensitivity

We fit $\log |P^c(i, i \pm r)|$ separately for reference rungs 16, 24 and 32, both directions, and windows 4–12, 8–20 and 12–28. Windows are clipped to endpoints inside rungs 9–56; fits need at least six points above 10^{-12} . Actual ranges and sign changes are saved. The threshold is an analysis filter, not a validated numerical accuracy bound. Power and exponential fits use the same points and two fitted parameters; their log residual RMS is only a descriptive comparison.

For the isolated rung reference the effective power slopes span 0.39–1.36, broader than the previously reported window-averaged fit near 0.76. The square paired *connected* slopes span 1.55–4.12, square stripe -0.50 –6.74, cubic stripe 0.37–15.55, and trellis stripe 1.92–13.64 for the displayed representatives. Negative effective slopes arise from fitting across spatial modulations. No common asymptotic exponent follows from these ranges. In particular, subtracting a paired-state plateau changes the function being fit.

This check follows the boundary concern demonstrated by Shen, Zhang and Qin, [Phys. Rev. B 108, 165113 \(2023\)](#): reference-bond selection can strongly affect extracted pair exponents. The source is used for the methodological motivation, not as a numerical benchmark for this extended, mean-field-coupled Hamiltonian. The supplied bulk-cut table additionally excludes 4, 8 or 12 rungs at each end. The representative square stripe’s long-distance magnitude remains 8.99 – $10.31 \cdot 10^{-5}$, cubic 1.15 – $1.39 \cdot 10^{-7}$ and trellis 6.19 – $6.97 \cdot 10^{-7}$ across those cuts. The finite-distance ordering survives this change, unlike any universal fitted-exponent claim. Independent larger- L/χ results remain necessary.

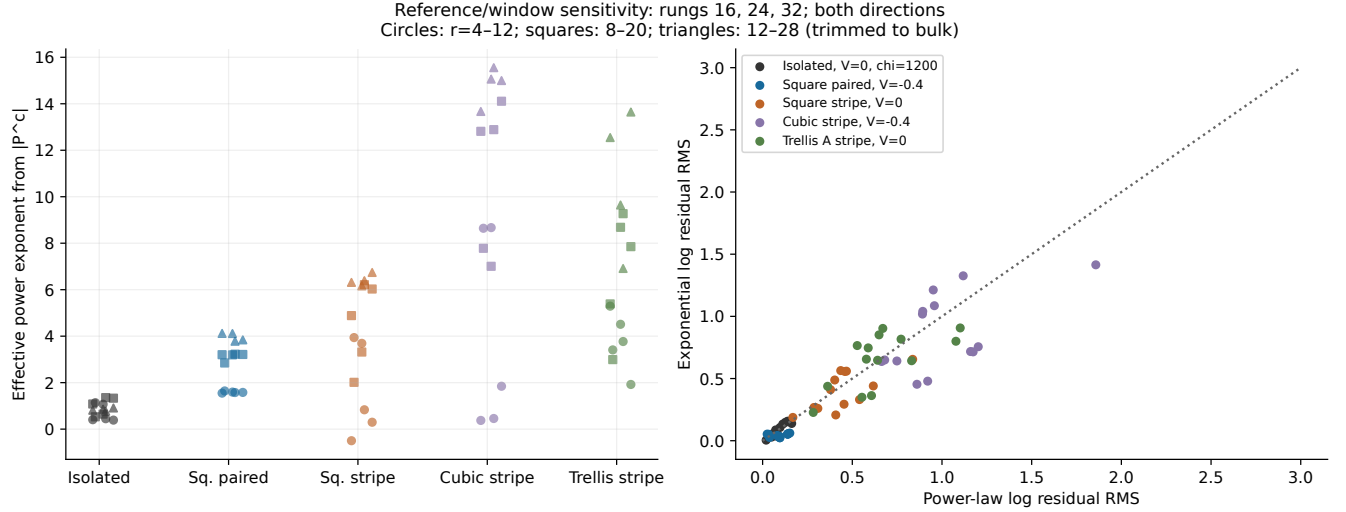


Figure 5: Fit sensitivity, not confidence intervals. The right panel compares two descriptive decay forms on identical observations. Neither a favorable R^2 nor a smaller residual certifies an asymptotic power law or exponential.

5 Completed square A/B campaign

All four runs at $t_0 = 1.4$ reach accepted fixed points. The analysis checks source/configuration/seed hashes, raw-update identities, final ten-step channel and density gates, inner-DMRG energy changes, slow-mode criteria, energy windows and Hamiltonian consistency. It also reconstructs canonical energies and their target-density correction with 256 physical sites per cell.

Table 2: Completed square A/B endpoints. Pair RMS is the historical physical symmetrized leg anomalous expectation on bonds 6–58, not D or a pair–pair correlator. The two seed energies agree; they are not distinct energy-ranked phases.

V	Sweeps	Seed	$E_{\text{target}}/(Nt)$	Leg-pair RMS
−0.4	40	Stripe	−1.167404800596	0.0352251
−0.4	40	Pairing	−1.167404800545	0.0352251
−0.2	55	Stripe	−0.914117842951	0.0335678
−0.2	44	Pairing	−0.914117843004	0.0335678

Within a fixed V , the seed energy difference is at most $5.3 \cdot 10^{-11} t/\text{site}$. Final bulk spin RMS is below $5.4 \cdot 10^{-7}$ at $V = -0.4$ and below $1.15 \cdot 10^{-6}$ at $V = -0.2$. Maximum A/B differences in physical leg-pair profiles are below $1.1 \cdot 10^{-6}$; residual charge differences reach $1.13 \cdot 10^{-5}$. The stripe displacement in the initial condition does not survive as a distinct stationary magnetic texture.

Connected long-distance rung-pair magnitudes are about $3.036 \cdot 10^{-4}$ at $V = -0.4$ and $2.475 \cdot 10^{-4}$ at $V = -0.2$, with nearly overlapping curves from both seeds and ladders. Full magnitudes remain about $1.166 \cdot 10^{-2}$ and $1.030 \cdot 10^{-2}$. The new cell reproduces the paired one-ladder snapshots’ correlations as well as their anomalous profiles. All measurements are *intraladder*; the mean-field spatial extension does not introduce interladder quantum entanglement.

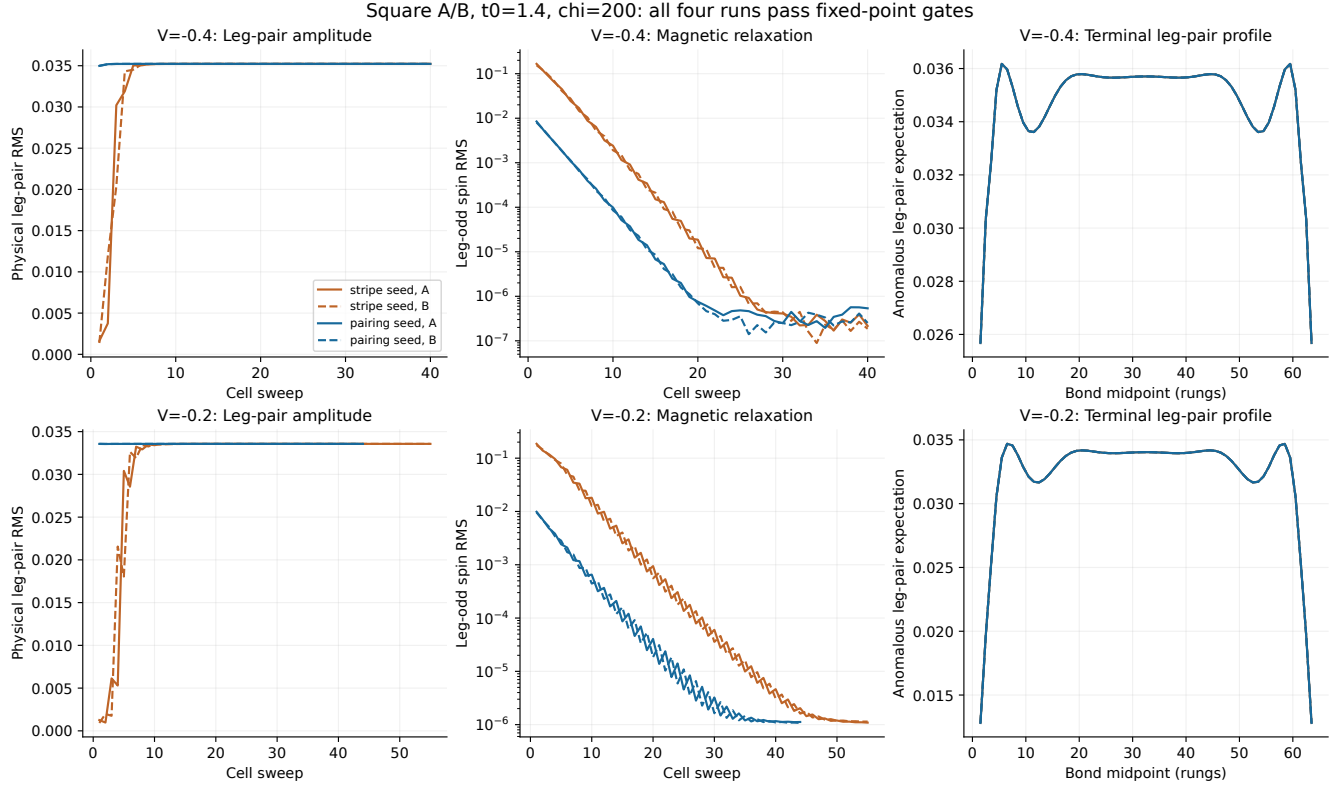


Figure 6: Both square seeds lose their magnetic component and approach the same paired solution at each attractive interaction. A/B curves nearly coincide. Unlike the retrospective stripe endpoints, these four states pass the stored stationarity gates.

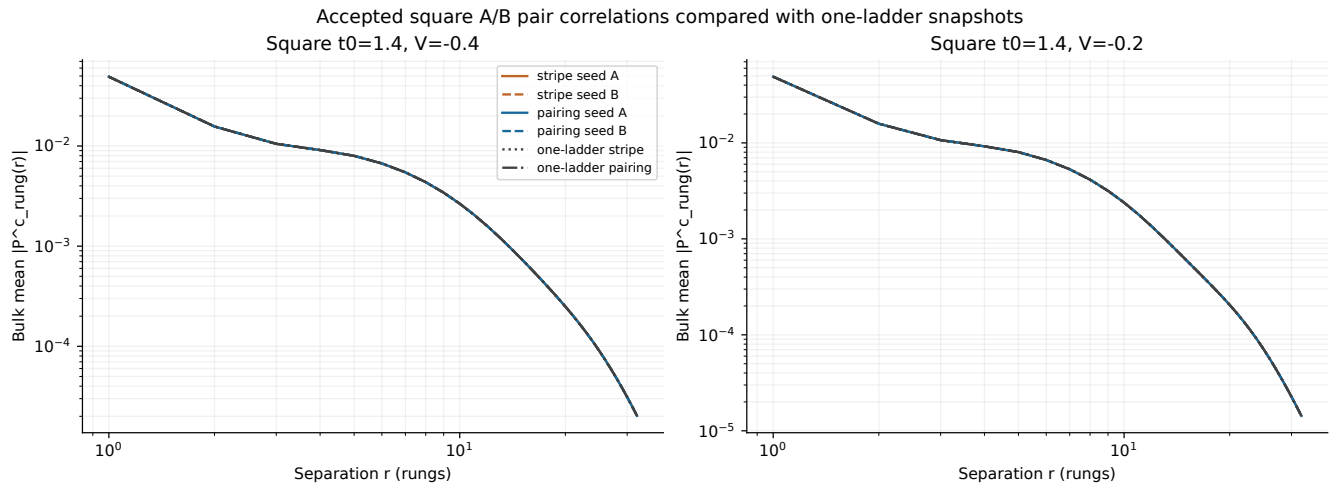


Figure 7: Connected rung correlations in accepted square A/B states and earlier one-ladder snapshots. Agreement supports the specific two-ladder extension, without establishing stability against all transverse modes.

6 Verification and reproducibility

Run `python scripts/analyze_pair_correlations_20260922.py` from the ladder project. It reads synchronized artifacts and writes this report's tables and vector/PDF and PNG figures. It never modifies states, manifests, receipts, histories, acceptance flags or measurement files; it runs no DMRG.

The source audit checks 66 complete version-2 diagnostic files against receipt hashes, full-source identifiers, compact-state hashes, model and configuration provenance. It checks finite matrices, Hermiticity, the four raw/connected Gram matrices' positive-semidefinite property to 10^{-9} , connected subtraction, within-class consistency and density/spin agreement with saved states. All measured imaginary parts and connected-subtraction residuals are zero in these real-state files. This establishes consistency of the supplied measurements, not independent convergence in L or χ .

- `coverage.csv`: original branch indices and retry provenance.
- `source_validation.json`: source paths, hashes and matrix checks.
- `correlation_summary.csv`: all 66 coupled MPSs and the isolated reference.
- `distance_curves.csv`: full/connected signed and magnitude curves.
- `reference_window_fits.csv`: every reference, direction and fit window.
- `bulk_cut_sensitivity.csv`: alternate edge exclusions.
- `stripe_wall_comparison.json`: both seeds and all striped trellis ladders.
- `square_AB_results.json`: four endpoint gate and energy audits.

The physical conclusion is finite-distance survival of local pairing in the measured stripes, together with suppressed long-distance propagation and surviving rung/leg sign structure. The new accepted square results strengthen the paired side of the comparison. Fully stationary stripes, matched bond-dimension/length checks and dynamical or phase-response observables are still required for stronger mechanism claims.